TECHNICAL SEO - SECTION 21

Crawl Budget and robots.txt

One file, four bytes of syntax, and the ability to remove your entire site from Google by accident.

> **Bottom line**
> robots.txt controls what bots may request. It is the highest-risk small file on your site: a 5xx response there causes Googlebot to stop crawling everything. Crawl budget, meanwhile, is the thing beginners worry about most and matters least until roughly 10,000 pages.

The technical SEO priority pyramid

Tier 3 works in this order, and the order is not negotiable. A failure at a lower layer invalidates the work above it.

1. **Crawlability and indexation** (Sections 21 to 22). The most common source of silent traffic loss.
2. **Canonicals and duplication** (Section 23).
3. **Redirects and status codes** (Section 24).
4. **Architecture** (Section 25).
5. **Rendering** (Section 26).
6. **Performance** (Sections 27 to 28).
7. **Structured data** (Section 30).

Fixing schema on a site whose pages are not indexed is polishing something invisible.

robots.txt: what it does and does not do

It requests that compliant bots not crawl certain paths. That is all.

It is not access control. It is voluntary. A bot can ignore it, and many do.

It does not remove pages from the index. This is the most consequential misunderstanding in technical SEO, and it deserves its own section below.

It must live at the root: `example.com/robots.txt`. Nowhere else.

The two rules that break sites

Rule one: robots.txt must return 200 OK.

If it returns a 5xx server error, Googlebot interprets that as "I cannot determine what I am allowed to crawl" and **stops crawling the site entirely** until it resolves. A 404 is fine and is treated as "crawl everything". A 500 is not. `[confirmed]`

Check it. Then check it again after any server change.

Rule two: never block CSS or JavaScript.

Google needs both to render the page as a user sees it. Blocking them means Google evaluates a broken version of your site. This was standard advice in 2010 and is now actively harmful. `[practitioner]`

The disallow plus noindex conflict

The single most common self-inflicted indexation bug, and it is worth understanding precisely.

You want a page out of the index, so you do both: `Disallow` it in robots.txt and add a `noindex` tag. That feels thorough. It is broken.

Disallowed means Google never fetches the page. Never fetching means never seeing the `noindex` tag. If the URL is linked from anywhere, Google can keep it indexed, showing the URL with no description, indefinitely. `[confirmed]`

Pick one:

- **To remove from the index:** allow crawling, use `noindex`. Google must be able to fetch the page to read the instruction.
- **To save crawl budget on pages you do not care about:** `Disallow`, and accept they might appear as bare URLs if linked.

For most sites, most of the time, the answer is `noindex` and leave robots.txt alone.

AI crawler directives

New in the last two years and now a deliberate decision rather than a default.

The critical distinction, covered fully in Section 40: **training crawlers and search crawlers are different bots.** Blocking indiscriminately opts you out of AI answers while trying to opt out of training.

A reasonable 2026 default:

``` User-agent: GPTBot Disallow: /

User-agent: ClaudeBot Disallow: /

User-agent: OAI-SearchBot Allow: /

User-agent: Claude-SearchBot Allow: /
```

User-agent: PerplexityBot Allow: /

User-agent: * Allow: /

Sitemap: https://example.com/sitemap.xml ```

Block the training bots, allow the retrieval bots that send referrals. Note that `anthropic-ai` **is a deprecated legacy agent** and including it in 2026 configurations produces broken instructions. `[practitioner]`

Remember rule one: robots.txt is voluntary. Real enforcement requires WAF or server-level rules, which are evaluated before robots.txt is even read. Section 40 covers that.

Crawl budget: mostly not your problem

Crawl budget is the number of URLs a bot will fetch from your site in a given window. It is determined by crawl rate limit (how fast your server responds without strain) and crawl demand (how much Google wants your content).

It becomes a genuine constraint above roughly 10,000 to 50,000 pages. `[practitioner]`

Below that, if pages are not being crawled, the cause is almost always architecture, internal linking, or quality, not budget. Optimizing crawl budget on a 200-page site is a way of feeling technical while achieving nothing.

When it does matter, the things that waste it:

- Faceted navigation generating infinite URL combinations
- Session IDs and tracking parameters creating unique URLs for identical content
- Endless pagination
- Large numbers of low-value auto-generated pages
- Slow server responses reducing the rate limit
- Redirect chains, since each hop costs a fetch

Reading the signals

In Search Console, **Settings, Crawl stats** shows requests over time, average response time, and what Googlebot spent its time fetching. Worth knowing it exists even if you rarely need it.

The useful reads:

- **Response time trending up** means your server is throttling the crawl rate.
- **Most requests going to non-HTML resources** means budget is being consumed by images or scripts.
- **Large numbers of 404s or redirects in the crawl** means wasted fetches.

Why this matters

this is the layer where problems are invisible from the front end. A site can look perfect, load fast, and read well, while a single line in robots.txt keeps a third of it out of the index. Nobody notices until traffic that never arrived is finally investigated.

Do this now

1. **Load `yourdomain.com/robots.txt` in a browser.** Confirm it returns content and a 200.
2. **Check for `Disallow: /`.** More sites than you would believe are blocking themselves, usually left over from a staging environment.
3. **Confirm CSS and JS are not blocked.**
4. **Look for any URL that is both disallowed and `noindex`.** Fix by picking one.
5. **Add your sitemap reference** if it is missing.
6. **Add the AI crawler block** above, adjusted to your preference. Decide deliberately whether you want to be in AI answers.
7. **Test in Search Console's robots.txt report** to confirm Google reads it as you intend.
8. **Open Settings, Crawl stats.** Note total requests, average response time, and any spike in 404s.
9. **Count your indexable pages.** Under 10,000, write "crawl budget is not my problem" and move on to Section 22.

Capstone step

Your robots.txt returns 200, blocks nothing it should not, contains no disallow-plus-noindex conflicts, references your sitemap, and states a deliberate AI crawler policy. You know whether crawl budget is a real constraint for your site or a distraction.

Key takeaways

- robots.txt returning a 5xx stops Googlebot crawling your entire site. It is the highest-risk small file you own.
- Disallow plus noindex is broken: disallowed means Google never reads the noindex, so the URL can stay indexed with no content. Pick one.
- Never block CSS or JavaScript. Google needs both to render what a user sees.
- Crawl budget only binds above roughly 10,000 pages. Below that, uncrawled pages are an architecture, linking, or quality problem.